

On Self-Supervised Learning

3. What SSL methods exist, and what is the principled reason we chose DINO over MAE, SimCLR, or iBOT?
4. What specific properties of DINO's learned representations make it more suitable than MIM for fine-grained classification?

DINO fits fine-grained transfer best because it learns semantic, local-global, negative-free representations better than MAE or SimCLR.

The evidence splits into three facets: the main **SSL families**, the principled case for **choosing DINO** over MAE, SimCLR, and iBOT, and the representation properties that make DINO stronger than **MIM** for fine-grained classification.

SSL Methods

SSL methods mainly group into **generative/predictive**, contrastive, and distillation or clustering-style approaches (Liu et al., 2020; Shurrab & Duwairi, 2021; Rani et al., 2023). Surveys also treat auxiliary pretext tasks as an earlier family and place methods such as SimCLR, MoCo, BYOL, SwAV, and DINO within the modern contrastive or self-distillation wave (Albelwi, 2022; Li, 2025; Ramesh et al., 2022).

- **Contrastive methods** learn by pulling augmented views together and pushing different images apart, typically with InfoNCE and mined positives and negatives (Jaiswal et al., 2020; Ramesh et al., 2022).
- **Generative or MIM methods** learn by reconstruction, and MAE is the main masked image modeling example in this comparison set (Liu et al., 2020; Moutakanni et al., 2025).
- **Distillation/prototype methods** such as DINO learn consistency in a teacher-student setup without labels, and some DINO derivatives add masked or patch-level objectives, as in iBOT (Caron et al., 2021; Govindarajan et al., 2024).

Why DINO

The principled reason to choose **DINO** over SimCLR is that DINO avoids negative pairs, so it sidesteps the false-negative or class-collision problem that contrastive methods inherit when semantically similar images are treated as negatives (Han et al., 2023; Zhang & Yu, 2022). The principled reason to choose DINO over MAE is that several comparisons attribute stronger downstream transfer to more semantic representations from contrastive or self-distillation teachers than from reconstruction-based MIM teachers (Kang et al., 2026; Boers et al., 2024; Han et al., 2026).

- DINO repeatedly shows **better transfer** than natural-image supervised pretraining and strong robustness after in-domain SSL pretraining (Boers et al., 2024).
- DINO is favored when domain labels are scarce and downstream tasks need **nearest-neighbor or retrieval** quality, not just reconstruction fidelity (Caron et al., 2021; Hussien et al., 2025).
- iBOT extends DINO with masked image modeling, but that supports a narrower conclusion: adding MIM can help, not that plain MIM is the better default for fine-grained transfer (Govindarajan et al., 2024).

Why DINO for Fine-Grained Tasks

DINO's learned representations appear more suitable than MIM for fine-grained classification because they preserve **semantic separability** while retaining useful spatial structure (Caron et al., 2021; Govindarajan et al., 2024). That combination matters in domains where classes differ by subtle morphology or texture, and the strongest task-specific comparisons show DINO widening its advantage as the classification problem becomes harder or labels become scarcer (Moutakanni et al., 2025).

- DINO features contain **explicit segmentation information** and rich spatial semantics, which supervised ViTs and convnets do not show as clearly (Caron et al., 2021).
- DINO is an **excellent k-NN embedding space**, indicating tight local neighborhoods that suit fine-grained retrieval and subclass discrimination (Caron et al., 2021; Govindarajan et al., 2024).
- MIM has a real strength in **attention diversity**, but evidence also says CL-based teachers have richer semantics, and DINO-like models outperform MAE on harder fine-grained tasks without finetuning (Kang et al., 2026; Moutakanni et al., 2025; Straka & Gruber, 2025).

Across these facets, the most defensible rationale is that DINO is the better default when the downstream goal is **fine-grained semantic discrimination under limited labels**, whereas MAE is stronger when reconstruction-style pretraining is the priority and iBOT is a useful DINO extension rather than a simpler first choice.

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